GROUP 189 – MAJOR PROJECT

CODEONE: DESIGN AND DEVELOPMENT OF A SECURITY-HARDENED COMPETITIVE PROGRAMMING ANALYTICS PLATFORM

1. Team Work Division (5 Members)

Divyanshi Kulshrestha (Team Leader & System Integration Developer): Managing project planning, coordinating team activities, maintaining the API-first modular monolith architecture and 15-table PostgreSQL/Prisma schema, building the smart problem tracker (user_problems state, bookmarks, personal notes) and daily/weekly/monthly goal management feature, and integrating completed modules.

Ishendra Rai (Platform Adapter & Synchronization Feature Developer): Designing the isolated PlatformAdapter interface, implementing policy-compliant connectors starting with the official Codeforces API, followed by AtCoder (via AtCoder Problems), Codewars, and a manual data import fallback for scraping-restricted platforms (LeetCode, CodeChef), alongside public-handle ownership verification and asynchronous sync_logs tracking.

Ayush Raj (Authentication, Security & Access Control Developer): Implementing Argon2id/bcrypt authentication, verified-email Google and GitHub OAuth 2.0, hybrid session lifecycle (15-minute JWT access tokens + 7-day rotating HttpOnly refresh tokens with token-family reuse detection in user_sessions and refresh_tokens), 4-role default-deny RBAC (User, Moderator, Admin, Auditor), IDOR protection, and append-only audit_logs.

Astha Goswami (Unified Analytics, Streak & Weakness Intelligence Developer): Developing the unified analytics dashboard, timezone-aware global streak calculator and 365-day activity heatmap derived from timestamped submissions, difficulty progression breakdown, and Level 1 rule-based topic weakness detection and daily problem recommendations.

Anvesha Singh (Contest Aggregator, Visualization & QA/Testing Developer): Developing the centralized contest calendar (CLIST.by ingestion with countdown timers and per-platform rating_history graphs), writing automated unit/integration/security tests (Jest, Supertest, mocked adapter fixtures, IDOR & token-reuse checks), and compiling analytics charts and verified documentation.

2. Project Execution Flow (Sequential Timeline)

Phase 1: Title Finalisation, API Policy Research & Work Distribution (Completed): Finalize the project title (CodeONE), audit external platform API documentation and Terms of Service (establishing the Permanent API Integration Policy), prepare initial project documentation, and assign module-wise responsibilities among the 5 team members.

Phase 2: Architecture Design, Repository Structure & Prisma Schema Setup (Completed): Establish the API-first modular monolith architecture, set up the client (React 18 + Vite) and server (Node.js 22+ + Express) structure, design the 15-table PostgreSQL schema via Prisma ORM (separating mutable user_problems state from immutable submissions history), and plan staged Redis caching.

Phase 3: Foundation, Hybrid Authentication & Security Controls (In Development): Implement email/password auth (Argon2id/bcrypt) and Google/GitHub OAuth 2.0, build 15-minute JWT access tokens with 7-day server-tracked rotating refresh tokens and reuse detection, and enforce 4-role default-deny RBAC, user_id-scoped IDOR protection, and append-only security audit logging.

Phase 4: Platform Adapter Layer, Codeforces Sync & Problem Management (In Development): Build the unified PlatformAdapter interface, implement public-handle ownership verification, integrate the official Codeforces API connector first (followed by AtCoder Problems, Codewars, and manual data import fallback), track asynchronous sync states via sync_logs, and build problem state tracking (status, notes, bookmarks) and coding goals.

Phase 5: Unified Analytics Engine, Streak Intelligence & Contest Calendar: Develop the unified dashboard, timezone-aware global streak calculator (with optional clearly labelled recovery mode) and 365-day contribution heatmap from submissions, separate per-platform rating trend graphs, Level 1 rule-based topic weakness detection, and CLIST.by contest calendar ingestion.

Phase 6: Security Verification, Adapter Contract Testing & Performance Tuning: Verify OWASP protections (Joi validation, parameterized Prisma queries, Helmet headers, CORS allowlist, auth/sync rate limiting), run Jest/Supertest suites (including mocked adapter contract tests and the 9-point security verification checklist), and profile PostgreSQL queries with EXPLAIN ANALYZE and optional Redis caching.

Phase 7: CI/CD Quality Gates, Deployment & Final Documentation: Configure Docker Compose and GitHub Actions CI workflows, prepare production deployment configurations (Vercel/Render or Nginx + Let's Encrypt TLS), and compile final project reports, verified benchmarks, and live demonstration artifacts.

3. Hardware & Software Requirements

Hardware Requirements

Category	Specifications
Computer/Laptop	Intel Core i5 or higher processor (or equivalent)
RAM	Minimum 8 GB (16 GB recommended)
Storage	256 GB SSD or sufficient free storage
Peripherals	Keyboard, mouse and monitor
Internet	Stable internet connection for external API synchronization

Software Requirements

Category	Specifications / Tools
Operating System	Windows 10/11, Linux (Ubuntu), or macOS
Programming Languages	JavaScript (ES6+), SQL, HTML5, CSS3 (with Joi, Argon2id/bcrypt, JWT, Passport)
IDE / Code Editor	Visual Studio Code
Application Frameworks	React.js (v18+), Vite, Node.js (v22+ LTS), Express.js, TanStack Query
Database & Caching	PostgreSQL (v15+), Prisma ORM, Redis (v7+, optional staged caching)
Visualization & UI	Recharts, Tailwind CSS, React Hook Form
Version Control & DevOps	Git, GitHub Actions (CI/CD), Docker, Nginx, Jest, Supertest

Note: This is a software-based web analytics and cybersecurity project. Physical sensor nodes or external hardware components are not required.